

# TE Curve Verification Pipeline Familywise ONNX Report - tree

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

## Overview

This report evaluates exported ONNX models from the dataset input-mode retraining program. Each dataset/input-mode section uses dataset-matched held-out test curves and lists the exact model artifacts loaded from `models/`.

The report is diagnostic and family-specific. It does not replace an official multi-index model-promotion decision.

## Output Artifacts

- output directory: `output/validation_checks/track2_familywise_onnx_report/tree/2026-07-07-14-58-29__track2_tree_familywise_onnx_report` ;
- summary YAML: `output/validation_checks/track2_familywise_onnx_report/tree/2026-07-07-14-58-29__track2_tree_familywise_onnx_report/track2_familywise_onnx_report_summary.yaml` ;
- model inventory CSV: `output/validation_checks/track2_familywise_onnx_report/tree/2026-07-07-14-58-29__track2_tree_familywise_onnx_report/model_inventory.csv` ;
- per-curve metrics CSV: `output/validation_checks/track2_familywise_onnx_report/tree/2026-07-07-14-58-29__track2_tree_familywise_onnx_report/per_curve_metrics.csv` .

## Simplified Dataset + Setpoints

- dataset: `simplified_dataset` ;
- input mode: `setpoints` ;
- evaluated family: `tree` ;
- dataset root: `data/simplified_dataset` .

### Models Used

| Surface  | Run Name                                         | Run Instance                                                         | Dataset Schema                   |
|----------|--------------------------------------------------|----------------------------------------------------------------------|----------------------------------|
| forward  | <code>te_tree_fw_simplified_setpoints</code>     | <code>2026-07-07-02-31-39_te_tree_fw_simplified_setpoints</code>     | <code>simplified_curve_v1</code> |
| backward | <code>te_tree_bw_simplified_setpoints</code>     | <code>2026-07-07-02-33-43_te_tree_bw_simplified_setpoints</code>     | <code>simplified_curve_v1</code> |
| global   | <code>te_tree_global_simplified_setpoints</code> | <code>2026-07-07-02-29-36_te_tree_global_simplified_setpoints</code> | <code>simplified_curve_v1</code> |

### Exact model paths

| Surface  | ONNX Model Path                                                                                                                               | Python Model Path                                                                                                                                   |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| forward  | <code>models/simplified_dataset/setpoints/exported/tree/forward/2026-07-07-02-31-39_te_tree_fw_simplified_setpoints/onnx/model.onnx</code>    | <code>models/simplified_dataset/setpoints/exported/tree/forward/2026-07-07-02-31-39_te_tree_fw_simplified_setpoints/python/tree_model.pkl</code>    |
| backward | <code>models/simplified_dataset/setpoints/exported/tree/backward/2026-07-07-02-33-43_te_tree_bw_simplified_setpoints/onnx/model.onnx</code>   | <code>models/simplified_dataset/setpoints/exported/tree/backward/2026-07-07-02-33-43_te_tree_bw_simplified_setpoints/python/tree_model.pkl</code>   |
| global   | <code>models/simplified_dataset/setpoints/exported/tree/global/2026-07-07-02-29-36_te_tree_global_simplified_setpoints/onnx/model.onnx</code> | <code>models/simplified_dataset/setpoints/exported/tree/global/2026-07-07-02-29-36_te_tree_global_simplified_setpoints/python/tree_model.pkl</code> |

### Aggregate Metrics

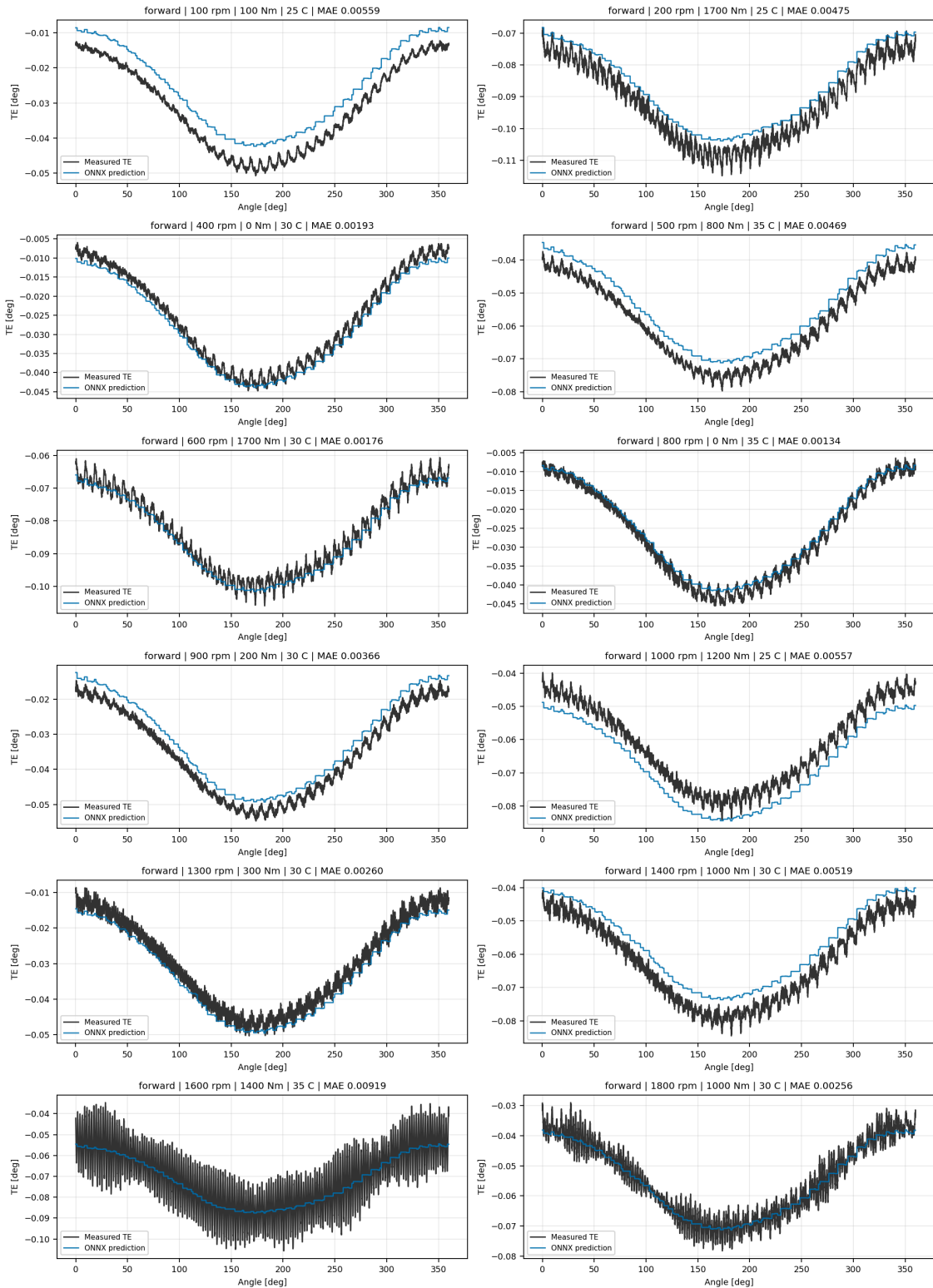
| Surface  | Curves | MAE[deg] | RMSE[deg] | Mean Error [%] | P95 Error[%] |
|----------|--------|----------|-----------|----------------|--------------|
| forward  | 97     | 0.003065 | 0.003435  | 6.740          | 11.934       |
| backward | 97     | 0.003345 | 0.003754  | 7.249          | 13.745       |
| global   | 194    | 0.003205 | 0.003594  | 6.995          | 13.161       |

### Offset And Shape Metrics

| Surface  | Signed Offset [deg] | Absolute Offset [deg] | Centered MAE [deg] | P2P Error [deg] |
|----------|---------------------|-----------------------|--------------------|-----------------|
| forward  | 0.000057            | 0.002559              | 0.001521           | 0.009543        |
| backward | 0.000218            | 0.002629              | 0.001776           | 0.009349        |
| global   | 0.000138            | 0.002594              | 0.001649           | 0.009446        |

## Forward 12-Curve Page

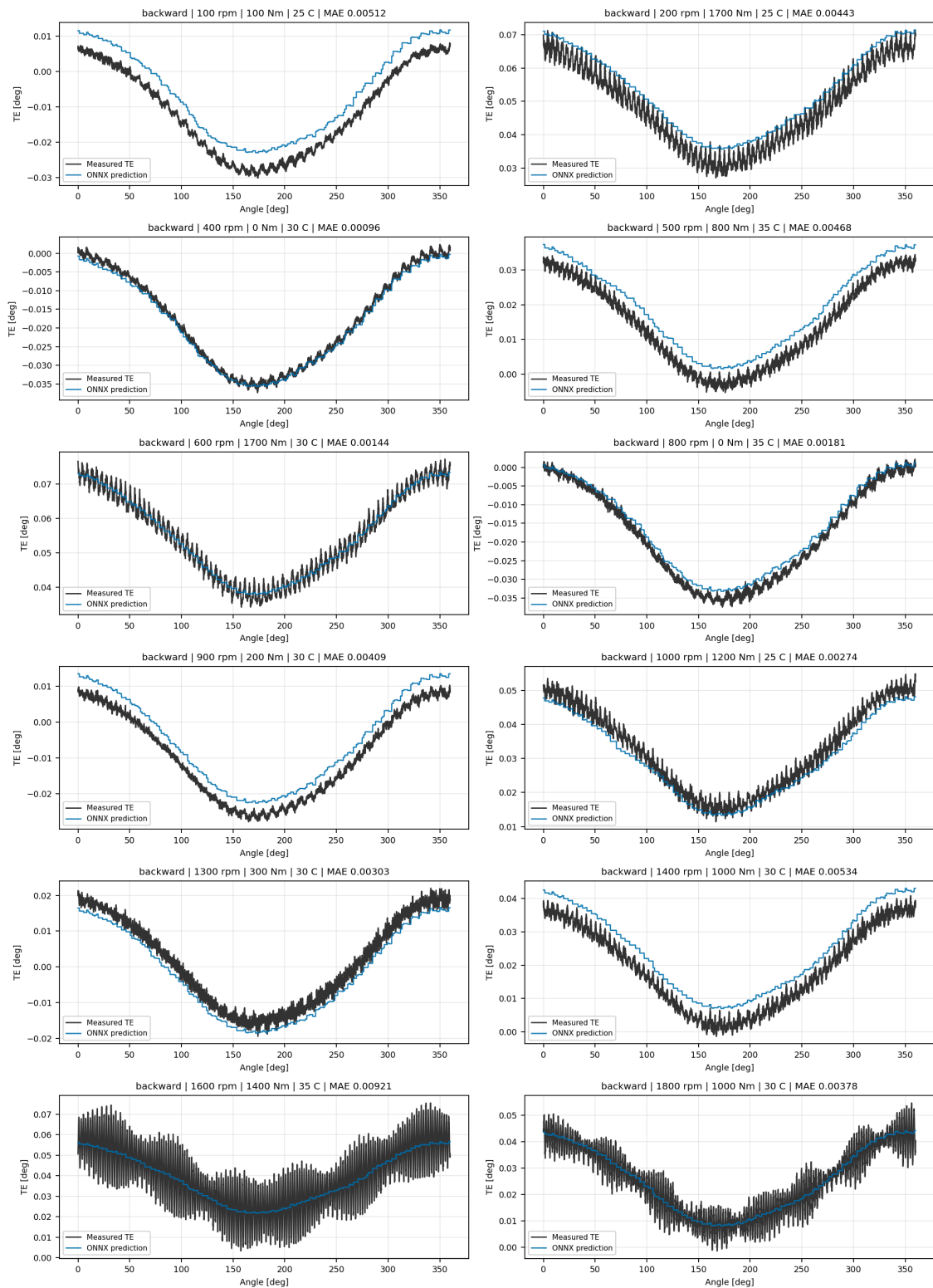
Simplified Dataset + Setpoints | tree | forward



simplified\_dataset\_setpoints forward 12-curve collage

## Backward 12-Curve Page

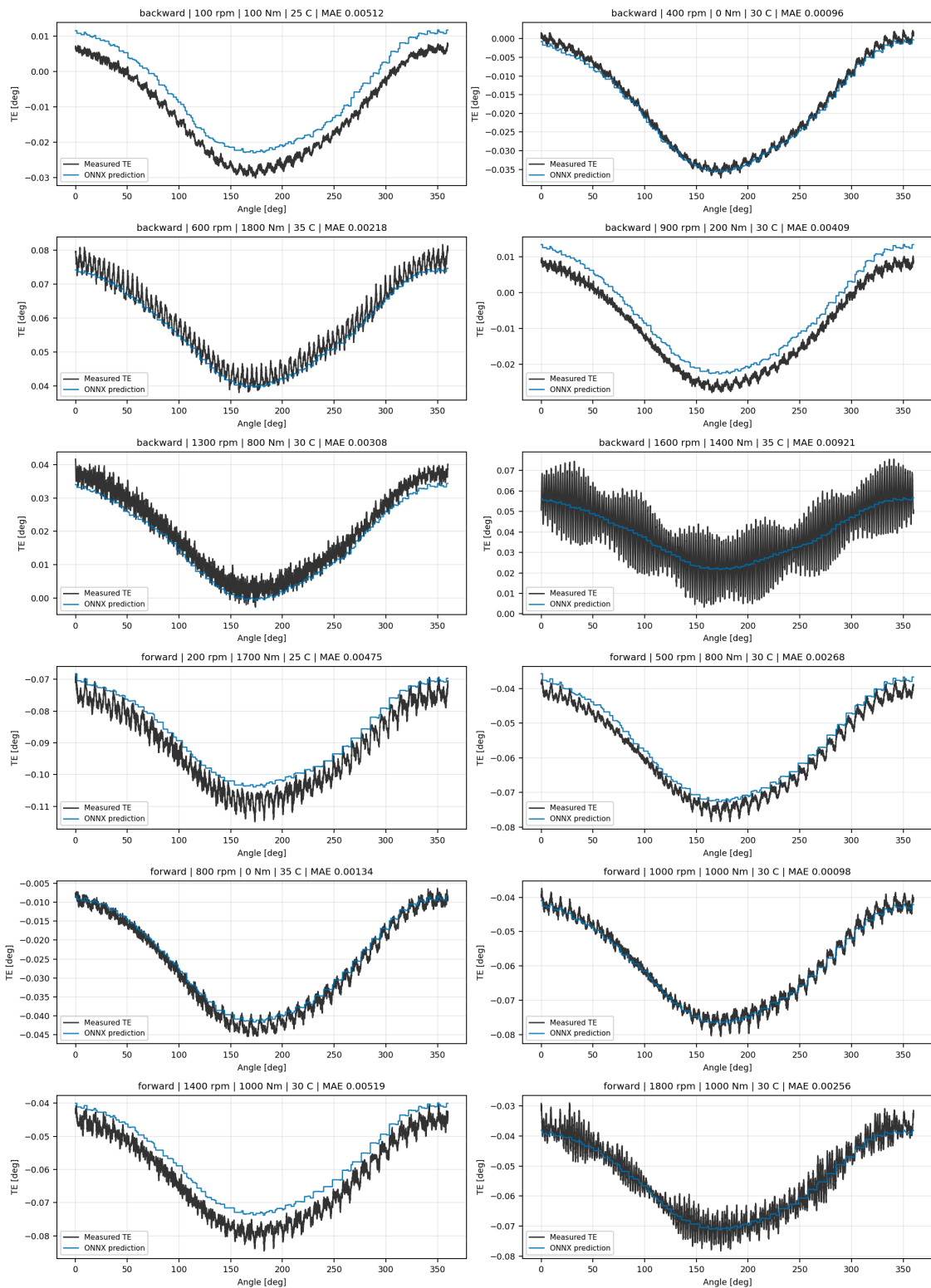
Simplified Dataset + Setpoints | tree | backward



simplified\_dataset\_setpoints backward 12-curve collage

## Global 12-Curve Page

Simplified Dataset + Setpoints | tree | global



simplified\_dataset\_setpoints global 12-curve collage

## Polished Dataset + Setpoints

- dataset: polished\_dataset ;
- input mode: setpoints ;
- evaluated family: tree ;
- dataset root: data/polished\_dataset .

### Models Used

| Surface  | Run Name                          | Run Instance                                          | Dataset Schema             |
|----------|-----------------------------------|-------------------------------------------------------|----------------------------|
| forward  | te_tree_fw_polished_setpoints     | 2026-07-07-09-34-40_te_tree_fw_polished_setpoints     | polished_setpoint_curve_v1 |
| backward | te_tree_bw_polished_setpoints     | 2026-07-07-09-37-16_te_tree_bw_polished_setpoints     | polished_setpoint_curve_v1 |
| global   | te_tree_global_polished_setpoints | 2026-07-07-09-31-48_te_tree_global_polished_setpoints | polished_setpoint_curve_v1 |

### Exact model paths

| Surface  | ONNX Model Path                                                                                                              | Python Model Path                                                                                                                  |
|----------|------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| forward  | models/polished_dataset/setpoints/exported/tree/forward/2026-07-07-09-34-40_te_tree_fw_polished_setpoints/onnx/model.onnx    | models/polished_dataset/setpoints/exported/tree/forward/2026-07-07-09-34-40_te_tree_fw_polished_setpoints/python/tree_model.pkl    |
| backward | models/polished_dataset/setpoints/exported/tree/backward/2026-07-07-09-37-16_te_tree_bw_polished_setpoints/onnx/model.onnx   | models/polished_dataset/setpoints/exported/tree/backward/2026-07-07-09-37-16_te_tree_bw_polished_setpoints/python/tree_model.pkl   |
| global   | models/polished_dataset/setpoints/exported/tree/global/2026-07-07-09-31-48_te_tree_global_polished_setpoints/onnx/model.onnx | models/polished_dataset/setpoints/exported/tree/global/2026-07-07-09-31-48_te_tree_global_polished_setpoints/python/tree_model.pkl |

### Aggregate Metrics

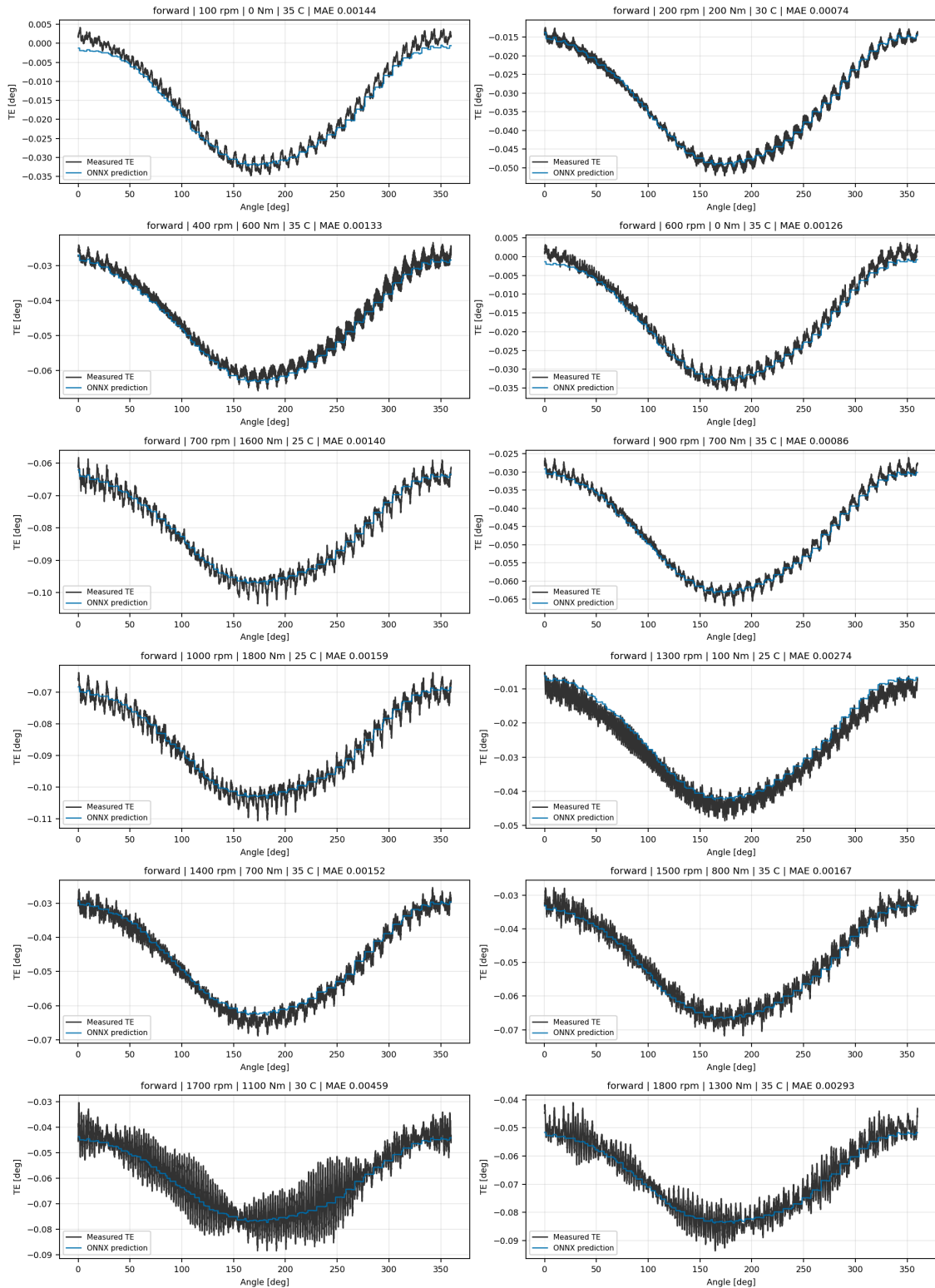
| Surface  | Curves | MAE[deg] | RMSE[deg] | Mean Error [%] | P95 Error[%] |
|----------|--------|----------|-----------|----------------|--------------|
| forward  | 100    | 0.002098 | 0.002536  | 4.314          | 8.754        |
| backward | 94     | 0.002684 | 0.003182  | 4.716          | 10.570       |
| global   | 194    | 0.002382 | 0.002849  | 4.509          | 10.152       |

### Offset And Shape Metrics

| Surface  | Signed Offset [deg] | Absolute Offset [deg] | Centered MAE [deg] | P2P Error [deg] |
|----------|---------------------|-----------------------|--------------------|-----------------|
| forward  | 0.000114            | 0.000909              | 0.001814           | 0.010853        |
| backward | 0.000289            | 0.000965              | 0.002319           | 0.012320        |
| global   | 0.000199            | 0.000936              | 0.002059           | 0.011564        |

## Forward 12-Curve Page

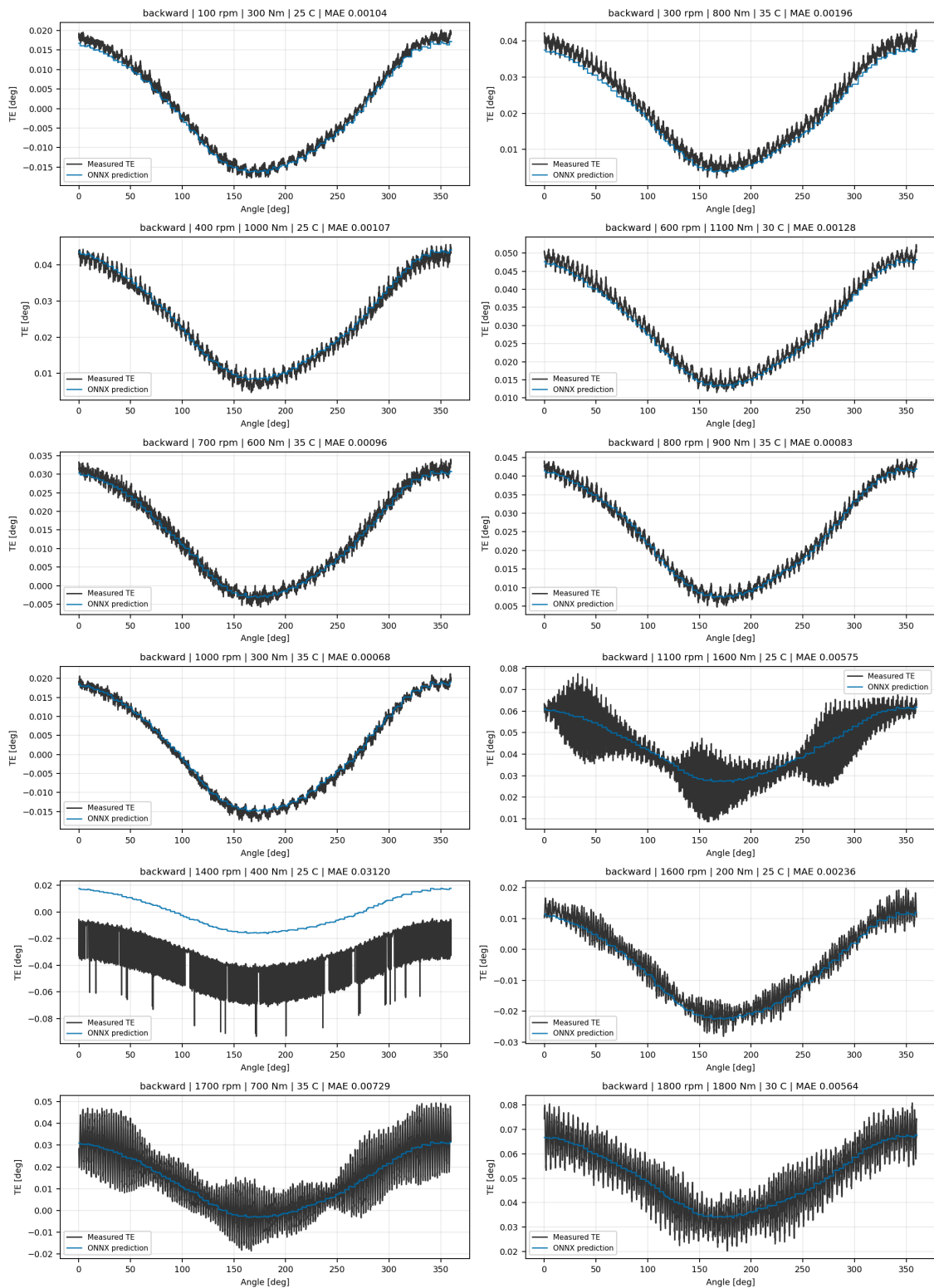
Polished Dataset + Setpoints | tree | forward



polished\_dataset\_setpoints forward 12-curve collage

## Backward 12-Curve Page

Polished Dataset + Setpoints | tree | backward

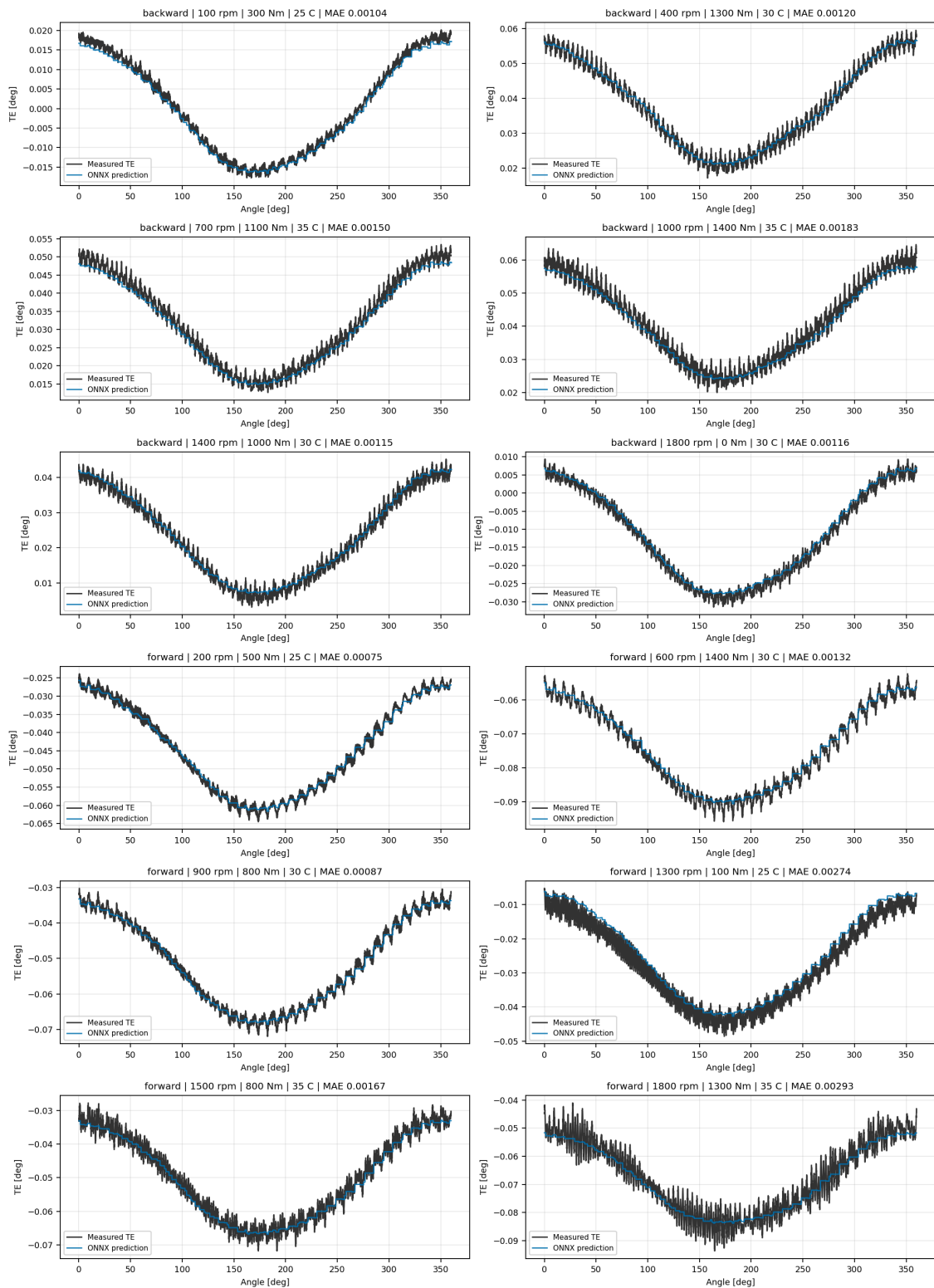


polished\_dataset\_setpoints backward 12-curve collage



## Global 12-Curve Page

Polished Dataset + Setpoints | tree | global



polished\_dataset\_setpoints global 12-curve collage

## Polished Dataset + Actual Values

- dataset: polished\_dataset ;
- input mode: actual\_values ;
- evaluated family: tree ;
- dataset root: data/polished\_dataset .

### Models Used

| Surface  | Run Name                              | Run Instance                                              | Dataset Schema    |
|----------|---------------------------------------|-----------------------------------------------------------|-------------------|
| forward  | te_tree_fw_polished_actual_values     | 2026-07-07-09-55-31_te_tree_fw_polished_actual_values     | polished_point_v1 |
| backward | te_tree_bw_polished_actual_values     | 2026-07-07-09-58-04_te_tree_bw_polished_actual_values     | polished_point_v1 |
| global   | te_tree_global_polished_actual_values | 2026-07-07-09-52-50_te_tree_global_polished_actual_values | polished_point_v1 |

### Exact model paths

| Surface  | ONNX Model Path                                                                                                                      | Python Model Path                                                                                                                          |
|----------|--------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| forward  | models/polished_dataset/actual_values/exported/tree/forward/2026-07-07-09-55-31_te_tree_fw_polished_actual_values/onnx/model.onnx    | models/polished_dataset/actual_values/exported/tree/forward/2026-07-07-09-55-31_te_tree_fw_polished_actual_values/python/tree_model.pkl    |
| backward | models/polished_dataset/actual_values/exported/tree/backward/2026-07-07-09-58-04_te_tree_bw_polished_actual_values/onnx/model.onnx   | models/polished_dataset/actual_values/exported/tree/backward/2026-07-07-09-58-04_te_tree_bw_polished_actual_values/python/tree_model.pkl   |
| global   | models/polished_dataset/actual_values/exported/tree/global/2026-07-07-09-52-50_te_tree_global_polished_actual_values/onnx/model.onnx | models/polished_dataset/actual_values/exported/tree/global/2026-07-07-09-52-50_te_tree_global_polished_actual_values/python/tree_model.pkl |

### Aggregate Metrics

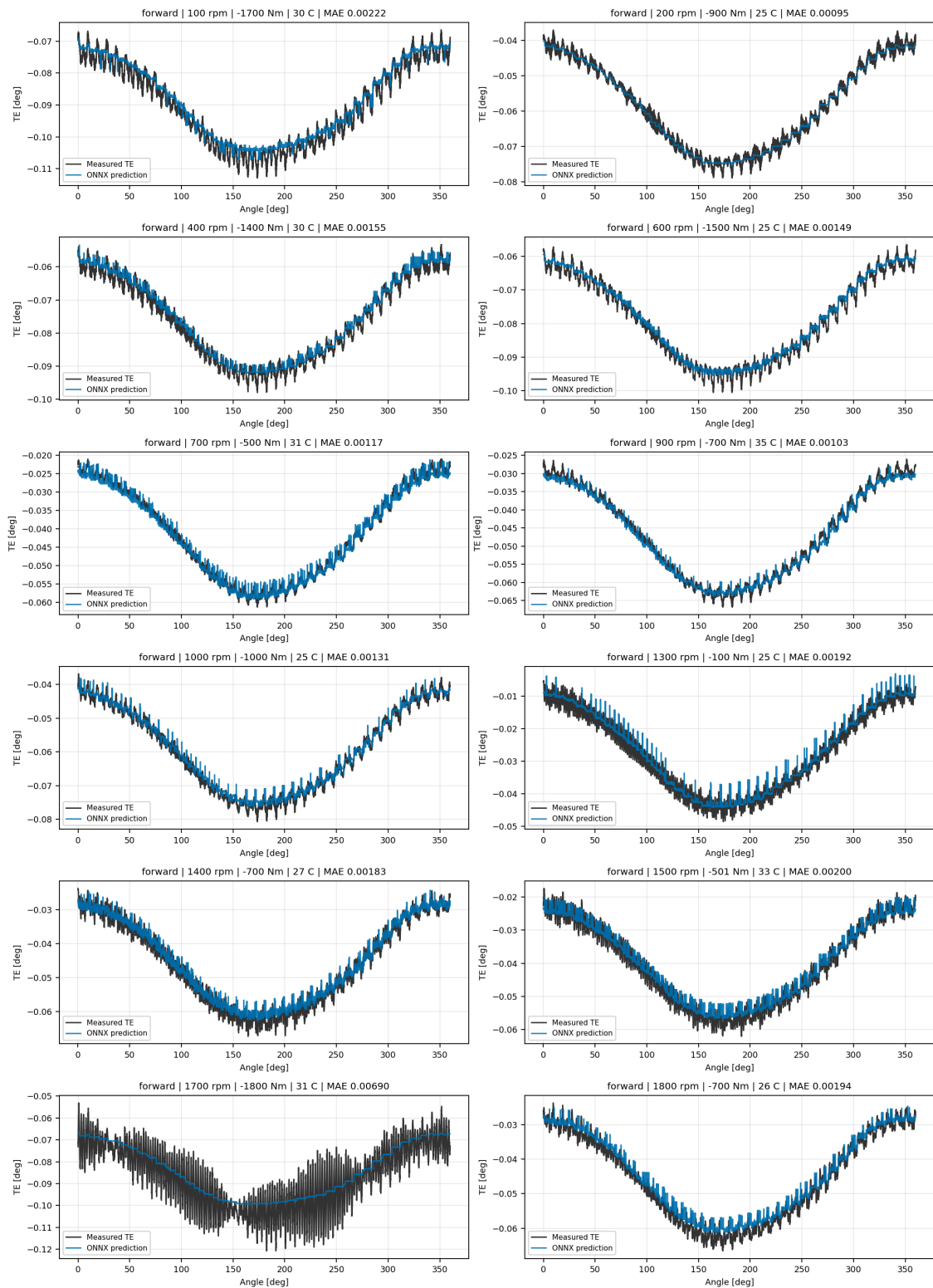
| Surface  | Curves | MAE[deg] | RMSE[deg] | Mean Error [%] | P95 Error[%] |
|----------|--------|----------|-----------|----------------|--------------|
| forward  | 100    | 0.002120 | 0.002599  | 4.341          | 8.662        |
| backward | 94     | 0.002759 | 0.003293  | 4.935          | 10.731       |
| global   | 194    | 0.002429 | 0.002936  | 4.629          | 9.989        |

### Offset And Shape Metrics

| Surface  | Signed Offset [deg] | Absolute Offset [deg] | Centered MAE [deg] | P2P Error [deg] |
|----------|---------------------|-----------------------|--------------------|-----------------|
| forward  | 0.000188            | 0.000799              | 0.001919           | 0.008543        |
| backward | 0.000294            | 0.000911              | 0.002406           | 0.009683        |
| global   | 0.000240            | 0.000853              | 0.002155           | 0.009096        |

## Forward 12-Curve Page

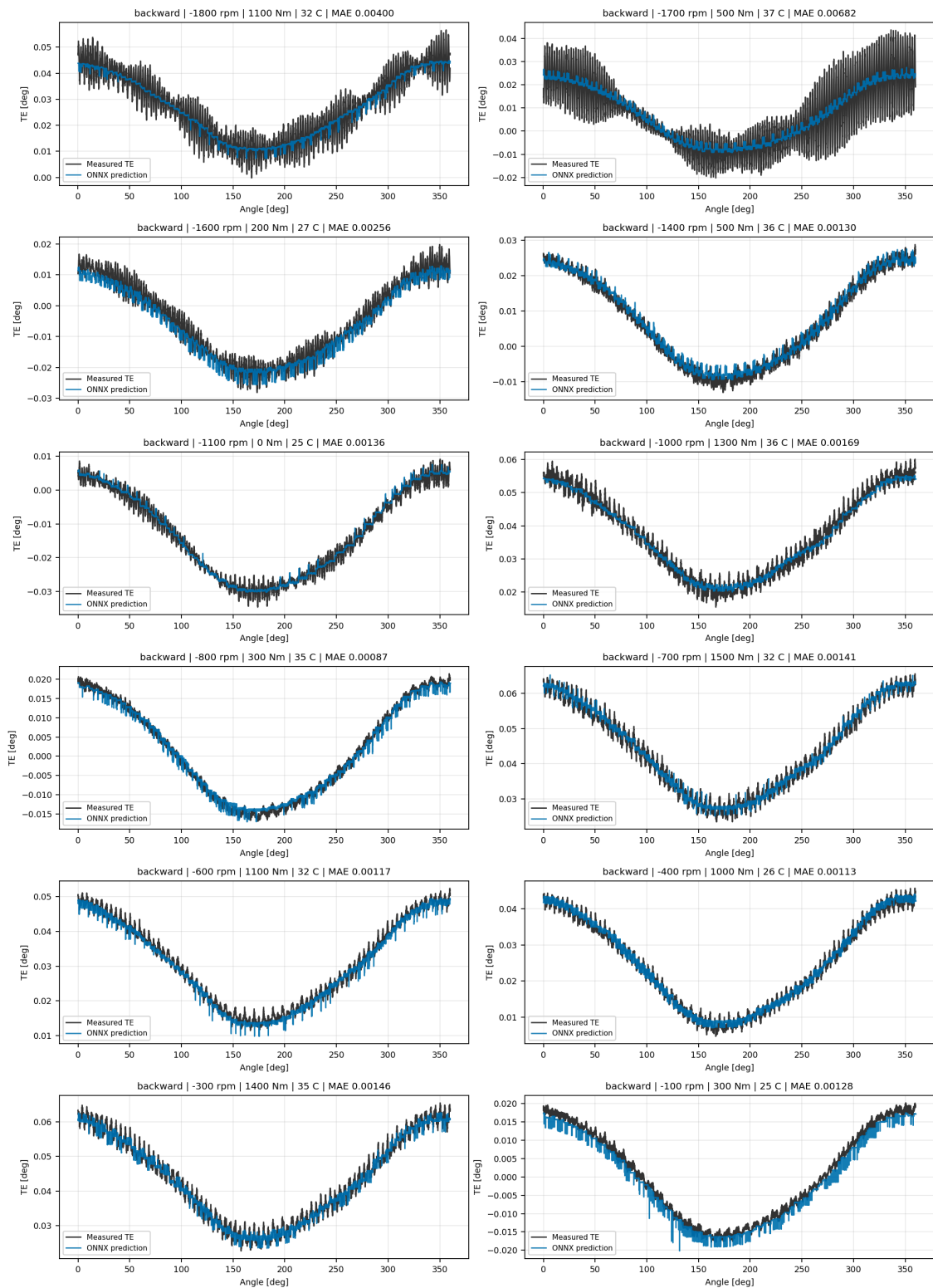
Polished Dataset + Actual Values | tree | forward



polished\_dataset\_actual\_values forward 12-curve collage

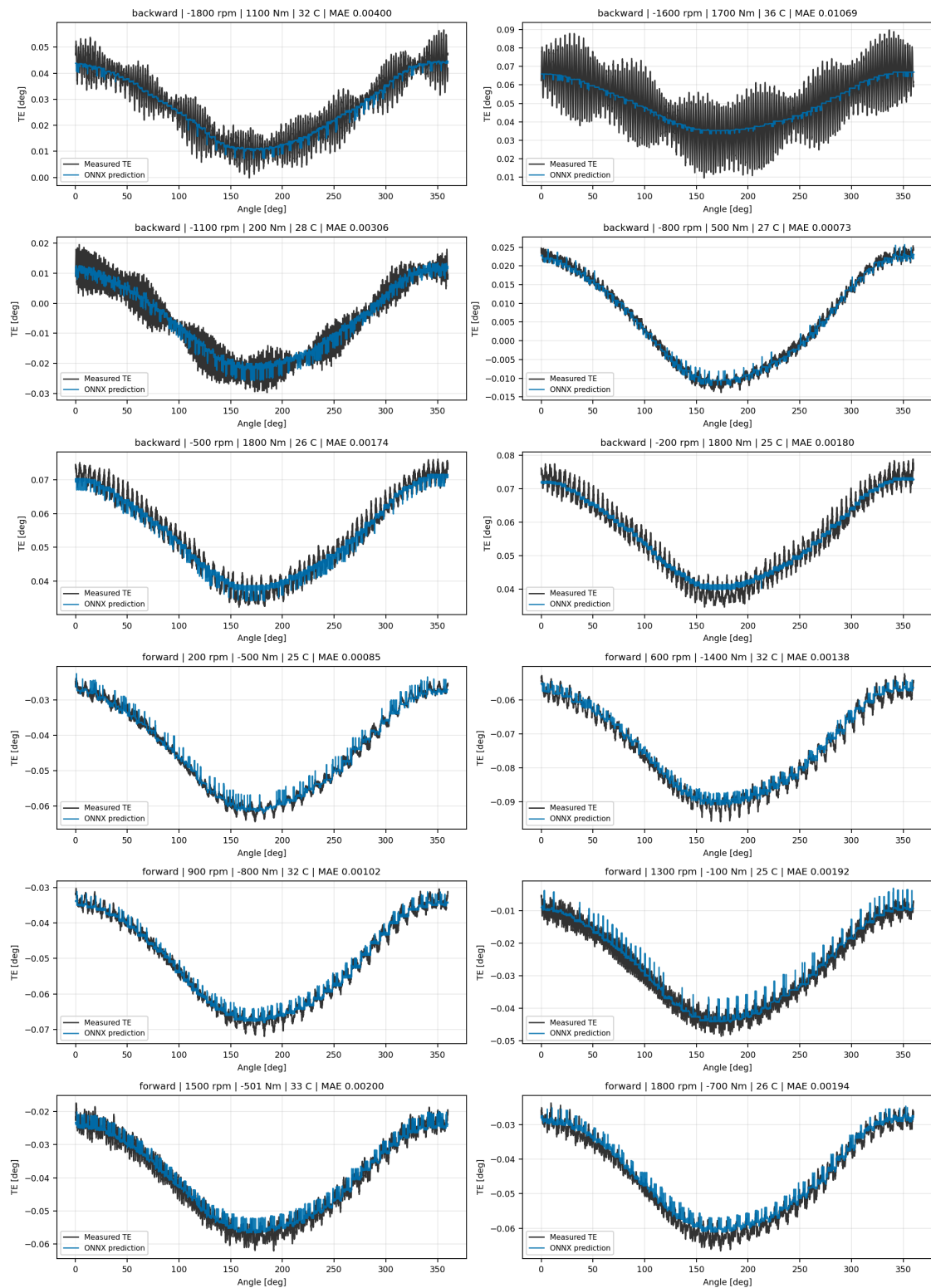
## Backward 12-Curve Page

Polished Dataset + Actual Values | tree | backward



polished\_dataset\_actual\_values backward 12-curve collage

## Polished Dataset + Actual Values | tree | global



polished\_dataset\_actual\_values global 12-curve collage